

What Is Science?

KEY QUESTIONS

- What are the goals of science?
- What procedures are at the core of scientific methodology?
- What is a scientific theory?

VOCABULARY

observation
inference
hypothesis
controlled experiment
independent variable
dependent variable
control group
data
theory

READING TOOL

List in order the parts of a typical experiment that uses scientific methodology. Use the headings in your text as a guide to fill in the chart in your  **Biology Foundations Workbook**.



Humans have wondered about their place in the cosmos since the earliest time. How do we fit into the grand scheme of nature? How much power do we hold over nature? What are the limits to our abilities? These and other questions are more relevant today than ever before. This chapter begins our effort to show you how science tries to answer those questions.

What Science Is and Is Not

This book will help you understand how biologists try to make sense of nature. These pages are filled with many scientific “facts” and ideas. But one of the first things you should understand is that scientific knowledge is always changing. Some “facts” and ideas you’ll find here may have changed since this text was written, and others will change soon. Why? Because scientists are constantly testing, debating, and revising scientific explanations of events in the natural world. That constant testing and revising helps explain why scientists don’t “believe” in scientific facts or ideas. Scientists either understand and accept a particular scientific explanation of the natural world, or they reject that explanation. But if science is not a list of unchanging facts and beliefs, what is it?

The Nature of Science The term *science* is usually defined as the use of evidence to construct testable explanations and predictions of natural phenomena, as well as the knowledge generated through this process. Most importantly, science is a process—an organized way of observing and asking questions about the natural world, developing those questions into testable explanations, and gathering and analyzing data that support or reject those explanations. The word *science* can also refer to the constantly growing and changing body of knowledge that the process of science generates.